SonoSPADE: Real-Time, Per-Tissue Ultrasound Texture Synthesis for Closed-Loop Acquisition

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Abstract. A closed-loop simulator for learning ultrasound *acquisition* renders a new B-mode frame at every agent step, so its texture stage must be fast and controllable. Realism-first diffusion is far too slow to run in the loop, and global sim-to-real translators use a single appearance map across the whole image, giving no per-tissue control. We present *SonoSPADE*, a per-tissue ultrasound texture stage that is real-time and GPU-free, trained unpaired with no real labels. The generator is conditioned on *both* a physics render and the tissue label slice. A segmenter, pretrained for free on aligned simulator pairs and then frozen, pseudo-labels the real pool for an OASIS-style discriminator and drives a per-tissue consistency loss. A single pass runs at 15 to 85 frames per second *without a GPU*, two to three orders of magnitude faster than diffusion. On the public Kaggle abdominal set, when we change a region's tissue label, only that region's texture changes, while methods that ignore the label do not react at all (label-swap locality ≈ 26, versus 0). Per-organ intensity matches real about $2\times$ better than any learned baseline, and structure is preserved $2.4\times$ better; on whole-frame FID/KID it trails by design. As a first in-the-loop proof of concept, on an organ-conditioned task scored by a *fixed* real-trained recognizer shared across texture stages, the per-tissue reward gives the highest ground-truth liver visibility in all three seeds. Code and the reproducible pipeline are released.[3]

Keywords: Ultrasound synthesis · Semantic image synthesis · Real-time synthesis · Closed-loop simulation · Sim-to-real.

1 Introduction

We study how to learn ultrasound *acquisition* in simulation: an agent moves a virtual probe and observes the B-mode frame (the standard grayscale ultrasound image) it would produce, so a texture stage must render a new frame at every step, and frames cannot be pre-rendered. The stage must be fast, must run on an ordinary computer, and must be controllable, so that the agent's actions map to anatomy. Realism-first synthesizers do not meet these needs. Diffusion produces high-quality ultrasound but takes seconds per frame [8], far too slow for a loop

[3] `https://github.com/tohnperfect/SonoSpade`

that uses millions of frames. Global sim-to-real translators work on the grayscale image alone [7], applying one appearance map to the whole frame, so they cannot make sure that liver reads as liver and kidney as kidney. Yet the pipeline already knows the anatomy: a canonical label slice drives a physics B-mode with acoustic shadowing, a depth-varying point spread function, and correlated speckle (the grainy interference pattern that gives each tissue its characteristic look). We make the learned stage *per tissue* by conditioning it on that label slice, and keep it single-pass so it runs in real time without a GPU.

What is new here is the combination. SPADE [5] and OASIS [9] synthesize from a label map alone and need real images with real labels. CUT [4] translates the grayscale frame with no label at all, while S-CycleGAN [7] and Vitale et al. [13] use segmentation only as a *loss* on a generator that never sees the label, so none of these three can be steered per tissue at inference (locality 0 for the label-free translators in Table 1). We instead *condition* the generator on the label *and* the physics render, and take the labels for free from a frozen simulator-pretrained segmenter: per-tissue steerable, no real annotation, and fast enough for the loop. Our contributions are:

- A real-time, GPU-free per-tissue texture stage: a single pass at 15–85 frames/s on a laptop with no discrete GPU, two to three orders of magnitude faster than diffusion, that runs in the loop without becoming the bottleneck.
- A physics-conditioned dual-input generator, conditioned on *both* the physics render and the label slice, giving per-tissue control (relabel a region and only its texture changes) with free unpaired segmentation consistency, so no real labels are needed.
- A per-tissue evaluation (per-class Wasserstein, a label-swap locality metric, speckle SNR, gray-level co-occurrence matrix (GLCM) texture), plus a diagnosis and two fixes (a sector remap and test-time adaptation) that move SonoSPADE from worst to ahead of the raw render on downstream transfer.
- A first in-the-loop proof of concept: under a *fixed* real-trained recognizer shared across texture stages, the per-tissue reward gives the highest ground-truth liver visibility in all three seeds.

2 Related Work

Semantic synthesis and label-consistent US. SPADE conditions generation on a label map via spatially-adaptive normalization [5], and OASIS recasts the critic as a per-pixel segmentation discriminator [9]; both are supervised and label-only. We borrow the adaptive conditioning but drive it from a physics render and supervise the discriminator with *free* pseudo-labels. A parallel line generates label-consistent US from anatomical models to train segmenters (e.g., a CycleGAN at 87–91 Dice on cardiac structures [1]); these whole-image translators offer no per-tissue control, which we expose by conditioning on the label.
Anatomy-consistent US synthesis. Closest to us, Vitale et al. [13] combine a segmentation-guided loss and polar (sector) training to suppress hallucinations and improve realism; related structure-preserving translators include

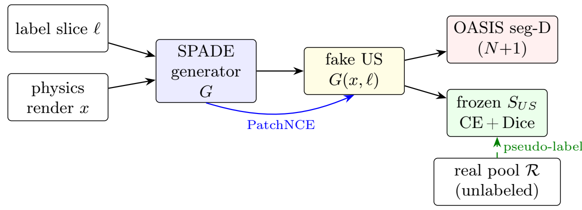

Fig. 1. SonoSPADE. The label slice and the physics render both condition the generator. A frozen segmenter S_{US}, pretrained free on aligned simulator pairs, pseudo-labels the real pool for an OASIS (N+1) discriminator and enforces a per-tissue consistency loss; PatchNCE preserves content. No inverse generator, no cycle loss.

S-CycleGAN [7,18] and CACTUSS [11]. These optimize offline realism; we optimize per-frame latency for in-the-loop use, and add per-tissue *conditioning* (not merely a loss), a label-swap locality metric, and single-pass GPU-free inference. Semantic diffusion [8] reaches higher fidelity but needs tens to hundreds of passes per frame, too slow for an agent that uses millions of frames.

3 Method

Fig. 1 summarizes SonoSPADE. Let x be a physics B-mode render and ℓ its one-hot canonical tissue label slice (K=13 classes). The generator $G(x, \ell)$ refines x into a realistic frame; it is trained unpaired against a real pool $\mathcal{R}$ with three forces.

3.1 Physics-conditioned dual-input generator

G is a ResNet encoder-decoder whose residual and upsampling blocks are SPADE-normalized: a parameter-free instance norm whose per-pixel scale and bias are predicted from the resized label map ℓ, so the semantic layout modulates activations at every scale [5]. The physics render x is the content stream, so G refines a tissue-correct render instead of hallucinating anatomy. Content is preserved by multilayer PatchNCE [4] (a patchwise contrastive loss tying each output patch to the input patch at the same location) on G's own encoder features between x and $G(x, \ell)$; there is no inverse generator and no cycle loss.

3.2 Free unpaired segmentation consistency

The simulator emits perfectly aligned (x, ℓ) pairs, so a compact U-Net [6] segmenter S_{US} is pretrained at no annotation cost, then *frozen* and used two ways. (i) It pseudo-labels the unlabeled real pool, giving an OASIS-style per-pixel discriminator D, which labels every pixel as one of the N grouped tissue classes of Sec. 3.3 or as *fake* (N+1 ways), a semantic target for free [9]: G wins only when each synthesized pixel reads as its conditioned tissue. (ii) A consistency loss $\mathcal{L}_{seg}$ = CE + Dice requires $S_{US}(G(x, \ell))$ to match ℓ, so G cannot collapse to a global recolor. The full objective is

$$\mathcal{L} = \mathcal{L}_{adv}^{\text{OASIS}} + \lambda_{nce}\mathcal{L}_{nce} + \lambda_{seg}\mathcal{L}_{seg} + \lambda_{tc}\mathcal{L}_{tc}, \tag{1}$$

where $\mathcal{L}_{adv}$ uses class-balanced per-pixel cross-entropy, so rare tissues are not ignored.

3.3 Grouped scheme and training

Fat, GI wall, pancreas, muscle, and generic soft tissue have near-equal backscatter and render almost identically, so S_{US} and D use a *grouped* scheme (these five merge, so K=13 canonical classes become N=9; distinct structures keep their class) while G still conditions on the full K classes. Three light stabilizers help: a temporal-consistency term $\mathcal{L}_{tc}$ requiring G to commute with a small in-plane shift and, on D only, an OASIS-style LabelMix [9,17] and an R1 penalty [3]: $\mathcal{L}_D = \mathcal{L}_{adv}^D + \lambda_{lm}\mathcal{L}_{lm} + \frac{\gamma}{2}\mathcal{L}_{R1}$, where $\mathcal{L}_{adv}^D$ is the same class-balanced cross-entropy on pseudo-labeled real and fake pixels. The deployed generator is an exponential moving average (EMA) of the training weights.
Hyperparameters. Every loss weight is 1 except λ_{seg}=5, and $\mathcal{L}_{seg}$ is an unweighted cross-entropy+Dice sum; Table S5 lists every weight, optimizer setting, and layer width.

4 Experiments

4.1 Setup

We build on an ultrasound substrate that runs on an ordinary laptop: it renders a physics B-mode from a CT-derived canonical label volume (13 tissues via TotalSegmentator [15]) and exposes a label-conditioned translator interface. Free aligned (x, ℓ) pairs come from eight abdominal CT cases with pose sweeps and domain randomization. For the real head-to-head, we use the public Kaggle abdominal ultrasound set [12]: every translator is trained on its 404-frame real train split and evaluated on the disjoint 60 annotated real frames (liver, kidney, gallbladder, spleen, vessels), a leakage-free real test on the same held-out content. Everything runs on Apple Silicon (MPS, no CUDA).

4.2 The label conditioning is genuinely per-tissue

The central claim is that texture is per-tissue, not a global recolor. Holding the physics render fixed, we relabel a tissue region to the reference tissue (liver) and measure the texture change inside versus outside. A global recolor scores near zero inside. The label-swap locality metric in Table 1 is a per-class summary: it averages each tissue's own inside/outside ratio, giving **26.3** for SonoSPADE versus 0 for every label-free translator by construction (supplementary Fig. S1, per-tissue values in Table S3). Per-class speckle SNR also shifts per tissue.

Reading the score. The score divides the mean absolute output change inside the relabeled region by the change outside it, so 0 means the output ignores the label, 1 means the edit spreads evenly over the frame, and the scale is unbounded above. Our 26.3 says the average tissue changes about 26 times more inside its own region than outside, from 10.4 (fat) to 60.7 (bone).

4.3 Free segmentation supervision

Pretrained on the aligned pairs, S_{US} reaches 0.93/0.90/0.72 Dice on liver/lung/vessels (acoustically distinct) on held-out simulated CT cases. The canonical 13-class macro Dice is capped at 0.50 by indistinct soft tissues; merging them in the grouped scheme lifts it to 0.66 (merged soft-tissue class 0.75 versus 0.35), with no change to distinct tissues.

Pseudo-label quality on real frames. We ran the same frozen checkpoint on the 60 annotated real frames, which it never saw. Macro Dice over the five annotated organs is 0.028 for the linear checkpoint that produced the reported pseudo-labels and 0.085 for the curvilinear retrain of Sec. 4.5 (liver 0.339 at 39 of 60 frames; kidney and spleen reach 0). Precision binds, since the masks annotate only five organs and any prediction on unannotated tissue counts against it. The pseudo-labels are a coarse regional prior for the discriminator rather than accurate anatomy, and the $\sim 3\times$ gap between checkpoints tracks the geometry mismatch of Sec. 4.5, not texture. The error enters at one point, the discriminator's real-branch target; the consistency term uses the exact simulator label and is unaffected (Table S4).

4.4 Comparison to baselines

Table 1 compares raw physics, CycleGAN [18], CUT [4], an S-CycleGAN-style variant [7], and SonoSPADE on the real Kaggle set. **Structure:** SonoSPADE preserves anatomy far better than any learned translator (edge IoU 0.41 versus 0.17/0.11 for CUT/CycleGAN, a $2.4\times$ margin): the label-free translators remap the frame to match appearance and drift on structure, while physics-plus-label conditioning keeps anatomy intact. **Per-organ realism:** We report intensity W_1 *restricted to the annotated organ masks*, since the whole-frame score is dominated by the background. Inside the masks, the recolors lose per-tissue fidelity (organ W_1 0.266/0.246 versus 0.064 for the render they started from), while SonoSPADE keeps it (0.119, about $2\times$ better than any learned baseline) and is the only method with per-tissue control (locality 26.3 versus 0). TSTR (train-on-synthetic, test-on-real: train a segmenter on translated frames, score it on real ones) is near zero for all linear methods (SonoSPADE lowest, 0.025), dominated by a linear-versus-curvilinear geometry gap and further depressed for SonoSPADE because its conditioning shifts features from what the classifier expects; matching the geometry and adding test-time adaptation reverses the ranking (next).

Table 1. Baseline comparison on the real Kaggle set. Edge IoU (structure vs. physics input, higher better); *organ* intensity W_1: Wasserstein-1 distance between output and real grayscale distributions inside each annotated organ mask, averaged over the four organs meeting the 200-pixel minimum (lower better); label-swap locality (0 for label-free translators by construction, higher better); TSTR Dice. SonoSPADE preserves structure 2.4× better and per-organ intensity about 2× better than any learned translator, and is the only method with per-tissue control. The SonoSPADE-curvi row is the same content in the matched sector geometry (Sec. 4.5); its TSTR is the five-seed mean from Table 2.

Method	Edge IoU ↑	Organ W_1 ↓	Locality ↑	TSTR Dice ↑
Raw physics	**1.00**	0.064	0.0	0.049
CycleGAN	0.11	0.266	0.0	0.032
CUT	0.17	0.246	0.0	0.031
S-CycleGAN (reimpl.)	0.13	0.223	0.0	0.026
SonoSPADE	0.41	**0.119**	**26.3**	0.025
SonoSPADE-curvi	0.54	0.115	18.6	0.099

Whole-frame realism (FID/KID). Fréchet and Kernel Inception Distance compare the distribution of whole frames to the real pool. SonoSPADE does *not* win whole-frame FID/KID (supplementary Table S1): its scores sit near the untranslated render, above the label-free translators. FID/KID score *overall* appearance, which a global tone map matches cheaply and which is blind to whether each organ reads correctly.

4.5 Ablation study

Ablating SonoSPADE's terms one at a time (retrained at a fixed budget; supplementary Table S2), the segmentation-consistency term is the largest single contributor: removing it roughly halves locality (27.1 to 13.7) and degrades structure (edge IoU 0.41 to 0.30) and per-organ W_1 (0.115 to 0.129). Locality does not fall to a label-free translator's zero (SPADE injects the label at every scale). Dropping the OASIS LabelMix or the temporal term leaves per-frame fidelity unchanged.

Downstream transfer: The near-zero TSTR in Table 1 is a geometry problem, not texture: our substrate renders a *linear* rectangular frame while Kaggle frames are *curvilinear* sector scans, so a segmenter trained on the linear render meets a different spatial prior at test time. Two drop-in fixes (Table 2). **(A)** A curvilinear remap warps the render (image and label) into the measured sector geometry (about 46°; supplementary Fig. S3) at train and test time, with probe-motion augmentation: TSTR rises about 4× for SonoSPADE and about 2× for the raw render, tying them (0.099 ± 0.011 versus 0.097 ± 0.006, 5 seeds). **(B)** Test-time adaptation (TENT [14]) of the segmenter's BatchNorm on 50 unlabeled real frames raises SonoSPADE to 0.124 but the raw render only to 0.106, so SonoSPADE *overtakes* on every one of the 5 seeds.

Table 2. Downstream transfer (TSTR Dice, higher better). Curvilinear and TENT rows are mean ± std over 5 seeds; the linear row is the single run from Table 1.

Setting	Raw physics ↑	SonoSPADE ↑
Linear render (Table 1)	0.049	0.025
+ curvilinear remap (A)	0.097 ±0.006	0.099 ±0.011
+ curvilinear + TENT (B)	0.106 ±0.006	**0.124** ±0.007

4.6 Inference latency: real time, no GPU

Because the stage runs inside a reinforcement-learning acquisition loop, per-frame latency, not peak realism, is the limiting factor. A single generator pass takes 66 ms on CPU (15 frames/s) and 12 ms on an integrated GPU (85 frames/s) at batch one, with *no discrete GPU* (`scripts/bench_latency.py`). A diffusion synthesizer instead runs T denoising passes per frame: a representative ADM/SDM-style denoising U-Net (the family Echo-from-noise [8] builds on; 62M parameters, sized to lower-bound the cost) costs 186 ms per pass on CPU, only about 2.7× a SonoSPADE pass, but its T passes make it two orders of magnitude slower *per frame* at T=50 (133× on CPU) and up to three at full DDPM. We verified this in place inside a MuJoCo [10] closed-loop acquisition environment, where the texture stage is not the bottleneck.

4.7 Per-tissue texture improves an in-the-loop acquisition agent

The latency result shows the texture stage *can* run in the loop; it does not show per-tissue texture *helps* the agent. That needs a task whose reward depends on per-tissue correctness, scored by a judge *fixed across* texture stages, not matched to each (the flaw that made our earlier goal-reaching benchmark uninformative, Sec. 5). We build one: *organ-conditioned acquisition.* A probe starts with the target organ partly in view and moves to bring it in; the per-step reward is the confidence of a frozen organ recognizer J that the organ is present. J is trained *only on real* Kaggle frames and masks (never on synthetic output, so the reward is non-circular) and is the *same* J for every texture stage. The main metric is texture-independent: the ground-truth fraction of the acquired view that is the target organ.

Is the reward usable? On held-out simulator content, J's liver IoU is 0.48 for SonoSPADE versus 0.21 (CUT) and 0.14 (CycleGAN), approaching the real-data ceiling 0.51 (Table 3, 3 seeds); the untranslated render scores 0.27, *above* both recolors, so part of the effect is structure preservation, not per-tissue texture alone. Trained under this reward (imitation-warm-started REINFORCE [16], 3 seeds, held-out CT cases), SonoSPADE reaches the highest ground-truth liver visibility, 0.152 ± 0.004, above CUT (0.144), CycleGAN (0.142), and the render (0.142) from a static start of 0.11, with all three per-seed differences positive and sign-consistent (directional, not significance-tested). Its reward signal is about

Table 3. Organ-conditioned liver acquisition (3 seeds, held-out CT cases). Top: a fixed real-trained recognizer J recognizes SonoSPADE's liver far better than a recolor's (real-data ceiling 0.51). Middle: trained under J as reward, SonoSPADE reaches the highest *ground-truth* liver visibility (static start 0.11), higher in 3/3 seeds.

	physics	CUT	CycleGAN	SonoSPADE
J liver recognition, IoU ↑	0.27	0.21	0.14	**0.48**
Ground-truth liver acquired ↑	0.142	0.144	0.142	**0.152**
Reward signal (J conf., secondary) ↑	0.20	0.18	0.16	**0.32**

twice as strong (0.32 versus 0.18–0.20); the ground-truth margin is *small* because the warm-start and liver's dominance already nearly solve the task.

The effect is confined to liver: for kidney, spleen, and gallbladder *no* translator (SonoSPADE included) yields a rendering the recognizer identifies (IoU ≈ 0), though the same J scores them 0.3–0.8 on *real* data. The bottleneck is the simulator and translator failing to reproduce their appearance, not the recognizer, so the multi-organ claim rests on the offline metrics and the in-the-loop gain is shown for the organ that transfers.

5 Limitations

Unpaired training matches per-tissue statistics, not exact speckle, so the output is training texture, not diagnostic imagery, and per-tissue fidelity is strong only for tissues acoustically distinct in B-mode. We validate on a *single* dataset and anatomy (abdominal Kaggle, 60 annotated real frames, 8 CT cases); other anatomies and scanners are untested. What is anatomy-specific is the 13-class label taxonomy and the fixed sector geometry, not the recipe: the method needs only a label volume and an unlabeled real pool, and no real annotations, and downstream Dice stays low even after the fixes, so transfer is directional, not clinical-grade. Our in-the-loop evidence is also bounded: it does not come from the earlier goal-reaching benchmark, whose per-texture *matched* judge is not comparable across texture stages (a random policy is not separated from the learned policies under that judge), and the organ-conditioned gain that replaces it is small, seed-consistent rather than significance-tested (3 seeds), and liver-only.

6 Conclusion

SonoSPADE is a real-time, GPU-free, per-tissue ultrasound texture stage for closed-loop acquisition: conditioning on both a physics render and the tissue label slice makes texture per-tissue and controllable, supervised for free by a frozen simulator-pretrained segmenter with no real labels. After matching the sector geometry it overtakes the untranslated render on downstream transfer, and its per-tissue reward gives an in-the-loop agent the highest ground-truth liver visibility under a fixed real-trained judge. More broadly, a fast, controllable texture stage makes closed-loop acquisition simulation practical, a step toward training autonomous and robotic ultrasound acquisition in simulation.

The motivating application is population-scale abdominal screening, where a deep model already assists diagnosis on real biliary-tract ultrasound [2]. Such systems assume a usable view has been acquired; learning acquisition in simulation targets that upstream step.

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